Label-efficient learning for High-Resolution Land Cover Classification using Large-Scale Self-Supervised Pre-Training and Transfer Learning

Dakota Hester

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Abstract

Land cover data is a key component of environmental analysis and informs decision making in agriculture, urban planning, hydrology, and other applied domains. Most publicly available land cover products are produced at a coarse resolution due to the inherent difficulty in classifying high resolution data. Deep learning semantic segmentation models have been shown to be performant for such tasks but require large amounts of fully annotated tiles for training. This is problematic for land cover classification due to the expense and difficulty of annotating such data at scale. A novel multi-stage pre-training approach is proposed to address this issue. In the first stage, a ResNet-152 model is pre-trained on 377,913 unlabeled images using a denoising autoencoder to enable spatial feature extraction without the need for labeled data. In the second stage, the pre-trained ResNet-152 model is used as an encoder in a U-Net architecture where fully supervised pre-training is performed using 10,220 labeled images from another region. Finally, in the third stage, the U-Net is fine-tuned with fewer than 500 labeled images over the state of Mississippi. A preliminary investigation of this approach using only 119 labels for training shows that the proposed approach outperforms the next best model by 4.57% in overall accuracy and 5.04% in overall F1 score, with a drastic 20.20% increase in macro-average F1 score. This approach has the potential to enable high-resolution land cover classification at scale while minimizing the need for human annotation to produce ground truth data, paving the way for high-resolution land analyses quickly and efficiently.

Contents

1 Introduction

Land cover data provides valuable information about the characteristics of the Earth's surface for a wide variety of applications across many domains, including agriculture, urban planning, and environmental monitoring [?, ?, ?, ?, ?]. Research into automated methods for land cover classification from remotely sensed data has been ongoing for several decades, with a majority of methods relying on supervised learning techniques to classify land cover using multispectral aerial or satellite imagery [?, ?]. Historically, pixel-wise classification methods have been the predominant approach for land cover classification, where each pixel in the image is classified based on the pixel's raw spectral values, handcrafted features, and ancillary data from other geospatial datasets [?, ?]. This pixel-wise paradigm is effective for classifying coarse and moderate spatial resolution imagery (e.g., MODIS, Landsat, Sentinel-2), but is less effective for classifying high spatial resolution imagery due to decreased inter-class spectral separability and increased intra-class spectral variability [?, ?, ?, ?, ?, ?]. This is problematic, as ideally, high spatial resolution (< 10 m) land cover data offers more detailed information about the Earth's surface with greater spatial accuracy, which is valuable for applications that require fine-grained information about land cover types and their spatial distribution [?, ?]. Object-based Image Analysis (OBIA) techniques have been proposed as a potential workflow to address these issues, but their performance is highly dependent on the quality of objects or superpixels extracted from the imagery during the segmentation process, which must be carefully tuned to the specific characteristics of the imagery and the land cover classes of interest [?, ?, ?]. Still, a key idea behind OBIA is that the spatial information contained in the objects can be leveraged to provide more information to the classifier, which addresses the inter-class spectral separability issues that hamper pixel-wise classification methods [?, ?, ?]. The application of deep computer vision models to land cover classification has been of significant interest in recent years, as these models excel at extracting spatial and spectral features from imagery in a data-driven manner, making them well-suited for high spatial resolution land cover classification tasks [?, ?, ?].

Convolutional neural networks (CNNs) are a class of deep learning models that have been widely used for land cover classification tasks due to their ability to automatically extract hierarchical spatial structures from the input imagery that are useful for classifying land cover type [?, ?, ?, ?]. Semantic segmentation models are a class of CNNs that consist of two main components: an encoder network that extracts spatial and spectral features from the input imagery, and a decoder network that generates a pixel-wise classification map from the features extracted by the encoder. These models learn a mapping from the input image to a pixel-wise classification probability map, which can be used to generate a land cover map by assigning each pixel to the class with the highest probability [?]. Semantic segmentation models have been widely used for land cover classification tasks, as they can facilitate the generation of high spatial resolution land cover maps that leverage both spectral and spatial information in the input imagery under a unified framework [?]. However, these models require large amounts of labeled training data to learn the complex spatial and spectral patterns present in the imagery, which can be difficult to obtain for high spatial resolution imagery due to the high cost of collecting and labeling the data [?].

This need for large amounts of ground truth data for accurate training of deep learning models has led to the extensive research into methods for reducing the amount of data necessary to train these models. Most of these methods focus on leveraging transfer learning, where the parameters of a model pre-trained using a large dataset are transferred to a model that is subsequently fine-tuned on a smaller dataset of interest. Transfer learning is effective for deep computer vision models, as the features learned by a model on a large dataset are often generalizable to other datasets thanks to the hierarchical nature of the features learned by the model [?, ?]. However, the effectiveness of transfer learning is highly dependent on the similarity between the source and target datasets, as well as the amount of labeled data available for the target dataset. For example, a common approach to transfer learning for land cover classification tasks is to pre-train a model on a large general-purpose image classification dataset (e.g., ImageNet [?]) and then fine-tune the model on a smaller dataset of interest [?]. This is generally effective, but the domain shift between the source and target datasets can hinder the performance of the model, and the extent of this domain shift between general purpose image classification datasets and high spatial resolution remote sensing datasets is not well understood [?]. Self-supervised pre-training methods have been proposed as an alternative to traditional fully supervised pre-training methods, as they can leverage the spatial and spectral information present in the input imagery to learn useful features through the use of pretext tasks that do not require labeled data [?, ?, ?]. These methods are effective, but most studies on the effectiveness of self-supervised learning are focused on single-stage pre-training methods where the only method used for pre-training is self-supervised learning, robbing the model of the benefits of supervised pre-training [?, ?, ?, ?].

This proposed research investigates the effectiveness of multi-stage pre-training methods for high spatial resolution land cover classification tasks to facilitate label-efficient training during the fine-tuning stage. The multi-stage approach addresses fundamental challenges in land cover classification: the substantial labeled data requirements for deep learning models and the domain shift between natural images (commonly used in pre-training) and aerial imagery. The proposed methodology comprises three strategically sequenced stages: self-supervised pre-training, supervised pre-training, and fine-tuning. In the initial self-supervised pre-training stage, a ResNet-152 encoder is trained on 377,913 unlabeled color-infrared imagery tiles covering Mississippi using a denoising autoencoder pretext task. This phase enables the encoder network to adapt from natural image features to aerial imagery characteristics without requiring manual annotations using a denoising task that forces the encoder to learn robust feature representations specific to aerial imagery patterns, textures, and objects, effectively addressing the domain shift problem by adapting the network to overhead imagery using abundant unlabeled data from the target region [?, ?].z During the supervised pre-training phase, the pre-trained ResNet-152 is incorporated into a U-Net architecture for training on 10,220 labeled tiles from the Chesapeake Bay Land Cover Dataset. This phase serves dual purposes: refining the encoder's feature representations to be specifically relevant for land cover classification rather than general image reconstruction, and initializing the decoder network parameters for generating pixel-wise classification maps. The supervised nature ensures the model learns discriminative features for distinguishing between land cover classes. In the final fine-tuning phase, the model is adapted to Mississippi's specific landscape characteristics using only 119 strategically sampled labeled tiles. This phase fine-tunes the model to the target domain, leveraging the knowledge learned from the previous stages to achieve superior performance with minimal labeled data, despite differences in the land cover classes and distribution between the Chesapeake Bay and Mississippi datasets. This progressive knowledge transfer approach is implemented with the goal of achieving superior performance with fewer labeled examples from the target domain, representing a potential order-of-magnitude reduction in manual annotation requirements for large-scale land cover mapping projects. Once this approach has been validated, it will be applied to the state of Mississippi to generate a 1 m resolution land cover map for the entire state, making it the highest resolution land cover map of Mississippi to date. A successful demonstration of this methodology will provide a blueprint for generating high spatial resolution land cover maps for other regions under label scarcity constraints, enabling cheaper and faster land cover mapping for a variety of applications.

2 Study area

The overarching goal of this proposed work is to develop a 1 m resolution land cover map for the state of Mississippi to facilitate high-fidelity land analysis. Mississippi is located in the southeastern United States, and is home to a diverse range of land cover types. The region's humid subtropical climate and fertile soils in the Mississippi River Delta make these areas home to a variety of agricultural activities [?], while the remainder of the state is characterized by forests and woodlands which comprise the majority of land cover in the state [?]. The coastal region along the Gulf of Mexico is home to a sprawling urban network intertwined with herbaceous wetlands and barren beaches [?]. The state's diverse land cover types, geographical position along the Mississippi River and Gulf of Mexico, and economic dependence on agriculture and natural resources make the need for accurate land cover information at high spatial resolution a priority in order to support natural resource management, agricultural planning, and environmental monitoring efforts.

National Agriculture Imagery Program (NAIP) For this study, USDA National Agriculture Imagery Program (NAIP) imagery is used as the primary data source from which the models are trained and the land cover map is generated. NAIP imagery is high-resolution (0.6 meter) multispectral aerial imagery collected during the summer months over the contiguous United States. Such imagery is collected on a 2–3 year cycle (depending on the state) and is predominantly used for agricultural purposes. While the temporal availability of NAIP imagery is limited, the high spatial resolution of the imagery and public availability of the data make it an attractive data source for remote sensing applications that require high spatial resolution data over large areas in the contiguous United States [?]. NAIP imagery collected in 2023 for the state of Mississippi is used in this study, which is the most recent data available at the time of writing. The imagery is provided in 4 spectral bands (Red, Green, Blue, and Near-Infrared) at a spatial resolution of 0.6 meters per pixel. The data is downloaded from the NRCS Geospatial Data Gateway, which provides color-infrared orthorectified, mosaiced imagery for each county for which NAIP imagery is available [?]. The imagery is resampled to a spatial resolution of 1 m to reduce the computational cost of processing and storing the data, as well as to align with the desired spatial resolution of the land cover product we aim to generate. Further, in accordance with recommendations from the Mississippi Automated Resource Information System (MARIS), the imagery is reprojected to the Mississippi Transverse Mercator (MSTM) coordinate system (EPSG: 3813) to ensure that all data is in the same coordinate reference system, as both UTM and State Plane coordinate systems divide the state of Mississippi into two zones, which can introduce complexity when working with datasets that span the entire state [?]. Figure ?? shows the extent of the study area and the NAIP imagery used in this study in the MSTM coordinate system.

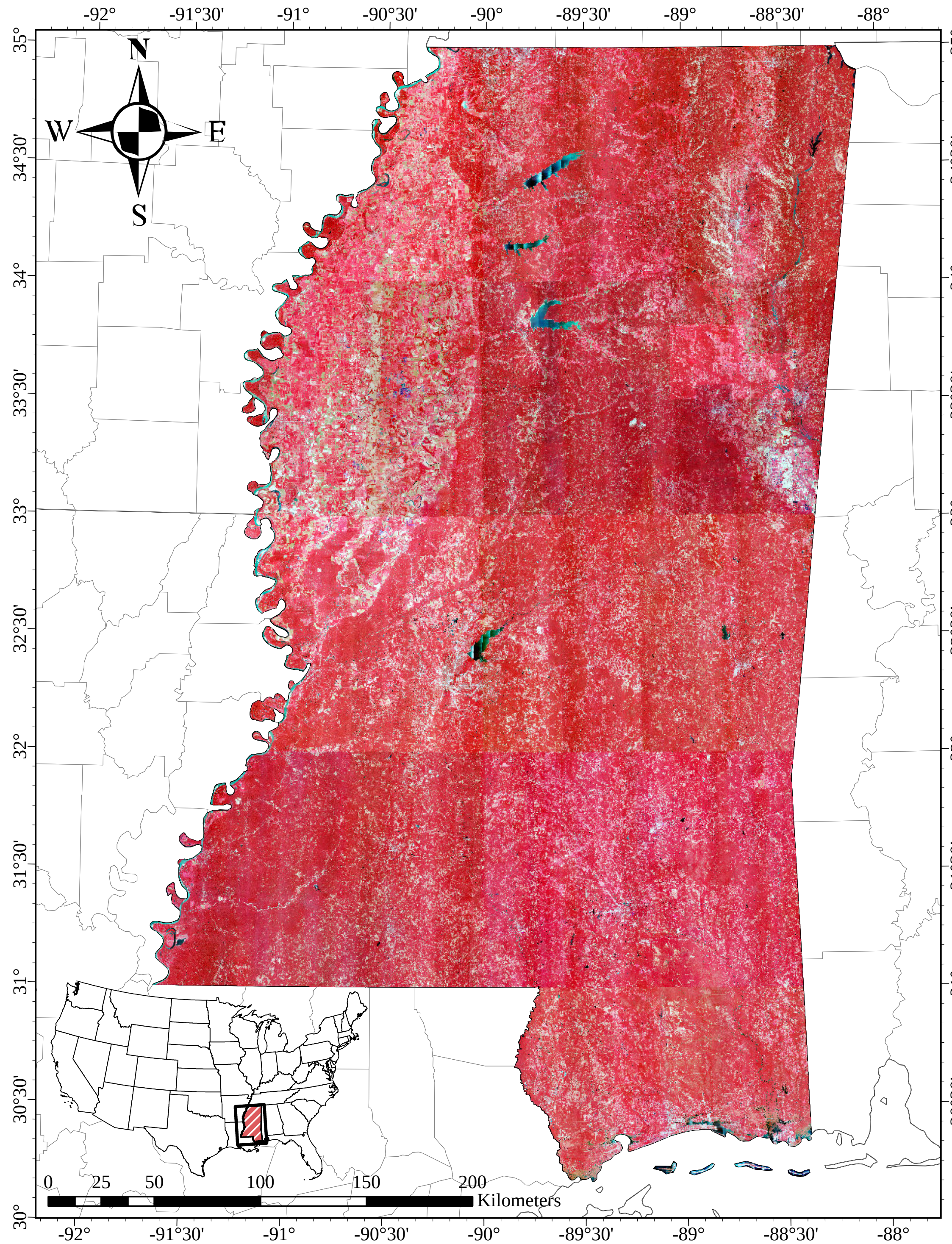

Figure 1: 2023 NAIP color-infrared imagery over Mississippi, USA.

3 Methodology

The proposed methodology consists of a multi-stage pipeline that leverages both self-supervised and supervised pre-training to develop a deep learning model for land cover classification over the state of Mississippi. Data for this proposed work consists of NAIP imagery over the state of Mississippi and Northern Virginia, as well as ground truth land cover data from the Chesapeake Bay Land Cover Dataset. For the Mississippi land cover tasks, a stratified sampling strategy is employed to select training, validation, and testing tiles for annotation that are representative of the study area using pre-existing coarse resolution land cover data to guide the stratification. The model architecture is based on U-Net, with a ResNet-152 encoder network and a decoder network that uses bilinear upsampling and residual connections to facilitate the flow of information through the network. The encoder of the model is pre-trained using a self-supervised denoising autoencoder approach, where a lightweight decoder network is used to reconstruct the input image from feature maps extracted by the encoder. The encoder network is then placed in a U-Net architecture and further pre-trained using supervised land cover data from Fairfax County, Virginia, before being fine-tuned on the Mississippi land cover data. Finally, the model is applied to the complete Mississippi NAIP dataset to generate a high-resolution land cover map for the state, and a thematic accuracy assessment is performed to evaluate the quality of the model's predictions.

3.1 Training data generation

Mississippi High Resolution Land Cover For this study, a subset of the Mississippi NAIP imagery is utilized for the purposes of training and evaluating the deep learning models for land cover classification. Sampled tiles are annotated by a team of analysts using the Computer Vision Annotation Tool (CVAT) platform. CVAT is an open-source web-based tool that facilitates rapid, high-quality, dense annotations of imagery data using a variety of utilities, including the ability to use a pre-trained Segment Anything Model (SAM) 2 [?] to quickly generate initial segmentation masks for objects using point prompts that can be refined and assigned to specific classes by human annotators [?]. The land cover legend used in this proposed methodology consist of 8 classes, with each class defined in Table **??**. The inclusion of an "Unclassified" class is intended to capture areas of the image that are poorly illuminated or occluded by shadows or other objects, where the annotators are unable to confidently assign a class label. This is particularly important due to the presence of dense forests in Mississippi, which cause varying intensities of shadows to be cast on the ground, which makes classifying the underlying land cover impossible in some cases.

National Land Cover Database (NLCD) Land cover data from the USGS National Land Cover Database (NLCD) is used to guide the stratification of the Mississippi NAIP imagery for the purposes of sampling training, validation, and test data. The NLCD is a multi-temporal land cover dataset produced by the Multi-Resolution Land Characteristics

(MRLC) Consortium, which is a partnership between the USGS, NASA, and other federal agencies. The newest edition of the NLCD, Annual NLCD, provides annual land cover data at a 30 m spatial resolution for the entire United States, and is derived from Landsat satellite imagery using a combination of spectral and temporal information. Thus, while the NLCD provides a lower spatial resolution than the NAIP imagery used in this study, it is valuable for providing a broad overview of land cover types across the state of Mississippi, and can be used to guide the stratification of the NAIP imagery for the purposes of sampling training, validation, and test data. Prior to use in this study, level 1 NLCD data is reprojected to the MSTM coordinate system to match the NAIP imagery, and is reclassified to match the land cover classes used in the Mississippi dataset as shown in Table **??**.

Table 2: Reclassification scheme for aligning NLCD land cover classes with our desired classification scheme.

NLCD Class Index	NLCD Class	Mississippi Class Index	Mississippi Class
11	Open Water	1	Open Water
23 24	Developed, Medium Intensity Developed, High Intensity	2	Impervious Structures
21 22	Developed, Open Space Developed, Low Intensity	3	Impervious Surfaces
31	Barren Land	4	Barren
41 42 43 90	Deciduous Forest Evergreen Forest Mixed Forest Woody Wetlands	5	Forest
52 71 81 95	Shrub/Scrub Grassland/Herbaceous Pasture/Hay Emergent Herbaceous Wetlands	6	Herbaceous
82	Cultivated Crops	7	Cultivated Crops
250	NoData	0	No Data

Chesapeake Bay Land Cover In addition to the NAIP imagery and the Mississippi land cover labels, auxiliary land cover data from the Chesapeake Bay Land Cover dataset is used during the supervised pre-training phase of the model. The 2022 edition of the dataset, which provides land cover information over the Chesapeake Bay watershed at 1 m ground sampling distance (GSD) for the 2017–2018 time period, is used in this study [**?**]. Specifically, land cover data over Fairfax and Arlington counties and portions of Alexandria, Loudon, and Prince William counties in Northern Virginia are used to guide the supervised pre-training phase of the model. This region was chosen due to the availability high-resolution land cover data, as well as the presence of dense urban and suburban land cover types that are not well-represented in the Mississippi NAIP dataset. Corresponding 2018 color-infrared NAIP imagery for this region is also used in this study, and is provided by the NRCS Geospatial Data Gateway. It is important to note that the orthoimages provided by the NRCS are not precisely clipped to the boundaries of the counties, and are instead provided as a single, large mosaic that includes multiple images with some overlap between adjacent counties. The land cover data is reprojected to the UTM Zone 18N coordinate system (EPSG:32618) to match the imagery, and the NAIP imagery is subsequently resampled to 1 m GSD such to align with the spatial resolution of the land cover data. Further, the land cover data is reclassified to more closely match the land cover classes used in the Mississippi dataset, as shown in Table **??**. The full extent of the subset of the Chesapeake Bay dataset used in this study is shown in Figure **??**. In total, 10,220 tiles are used for the supervised pre-training phase of the model, while 5,110 tiles are used for validation, and 5,109 tiles are used for testing.

Table 3: Reclassification scheme for reducing Chesapeake Bay land cover classes to more closely align with our desired classification scheme.

Chesapeake Class Index	Chesapeake Class	Reduced Class Index	Reduced Class Label
1	Water	1	Open Water
3	Tree Canopy	2	Forest
10	Tree Canopy over Impervious Structures		
11	Tree Canopy over Other Impervious		
12	Tree Canopy over Roads		
4	Shrubland	3	Herbaceous
2	Emergent Wetlands	4	Herbaceous
5	Low Vegetation		
6	Barren Land	5	Barren
7	Impervious Structures	6	Impervious Structures
8	Other Impervious	7	Impervious Surfaces
9	Roads		

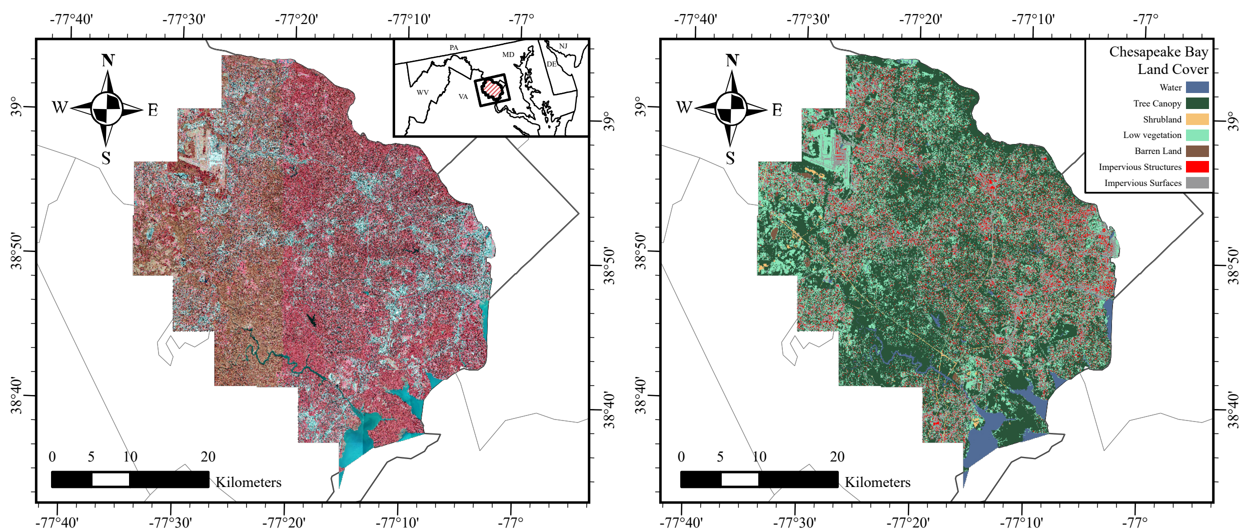

Figure 2: Extent of 2018 NAIP imagery (left) and 2018 Chesapeake Bay land cover (right) over Alexandria, Arlington, Fairfax, Loudon, and Prince William counties in Virginia, USA used in this study.

3.2 Stratifying by land cover composition

Ideally, land cover classification models should be trained and validated against samples that are representative of the study area. However, under the constraints of limited labeled data, it may be difficult to capture the full range of variation in the study area. Stratified sampling is a common approach to address this issue, where the study area is divided into strata based on some characteristic (such as land cover class or sub-region) and samples are drawn from each stratum. Unlike traditional land cover classification models, semantic segmentation models necessitate the use of densely annotated tiles that exhibit a high degree of diversity with regards to spatial characteristics. Ideally, a single training tile should consist of a mix of land cover classes, and the arrangement of land cover classes should similarly be representative of the study area as a whole. To achieve this, a stratified sampling strategy is employed using the NLCD to guide the selection of training, validation, and testing tiles. This ensures that both the distribution of classes in the tiles is representative of the study area and the content of the tiles is heterogeneous and diverse in terms of spatial characteristics.

A grid of 256×256 m square polygons is overlaid over the study area to create candidate tiles, and the proportion of each land cover class is calculated for each tile by computing zonal statistics on the NLCD data within each tile. Once the relative frequency of each land cover class is calculated, these frequencies are scaled by subtracting the median value and dividing by the interquartile range of each class. Principal component analysis (PCA) is applied to the scaled land cover relative frequency data to reduce the dimensionality of the relative prior to clustering. Once the principal components are calculated, the first two principal components are identified as capturing 94.24% of the data variance; subsequently, these principal components are used to guide the stratification of the data. Figure **??** visualizes these first two principal components of the land cover data at the tile level. Higher values of the first principal component correspond to a higher proportion of impervious land cover types in each tile, while higher values of the second principal component correspond to a higher proportion of forested and herbaceous land cover, with lower values correlating with regions that consist of tiles primarily or wholly composed of cultivated crops. These tiles are subsequently grouped into 250 clusters using K-means clustering operating on these two principal components. As shown in Figure **??**, the clustering algorithm effectively divides the principal component space into regions that correspond to differing land cover compositions. From each cluster, 2 candidate tiles are sampled for training, 1 for validation, and 1 for testing. This results in a total of 500 training tiles, 250 validation tiles, and 250 testing tiles. From the remaining candidate patches, 20% of the tiles are randomly sampled for pre-training and 5% are sampled for pre-training validation, yielding 377,913 and 94,480 tiles, respectively.

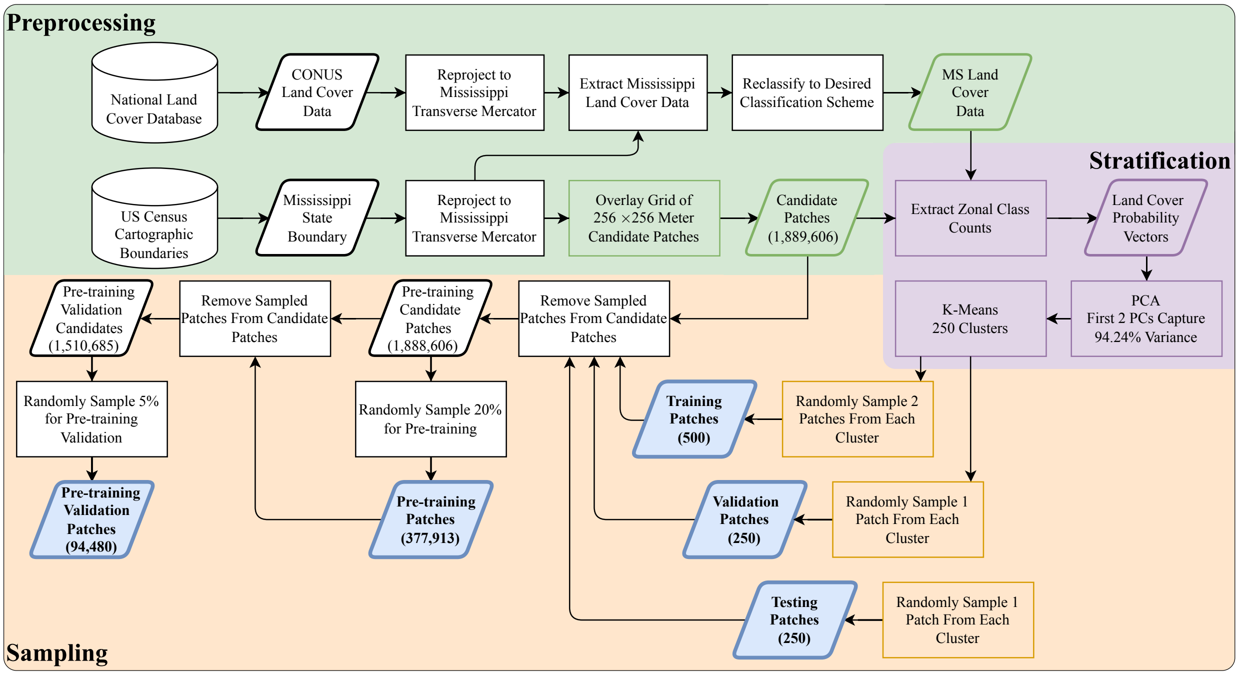

Figure 3: Flowchart of the stratified sampling process for selecting data for model training and validation.

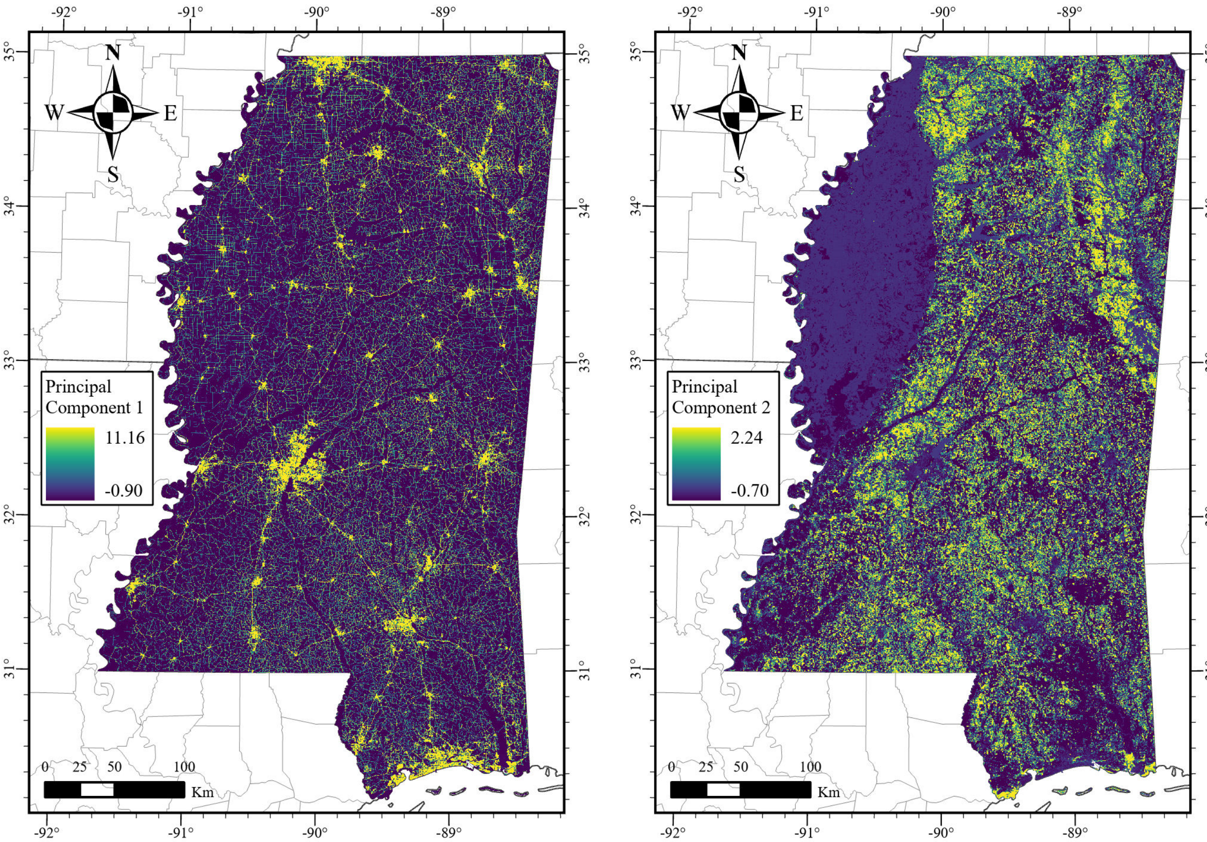

Figure 4: Principal components of NLCD relative frequencies per candidate tile.

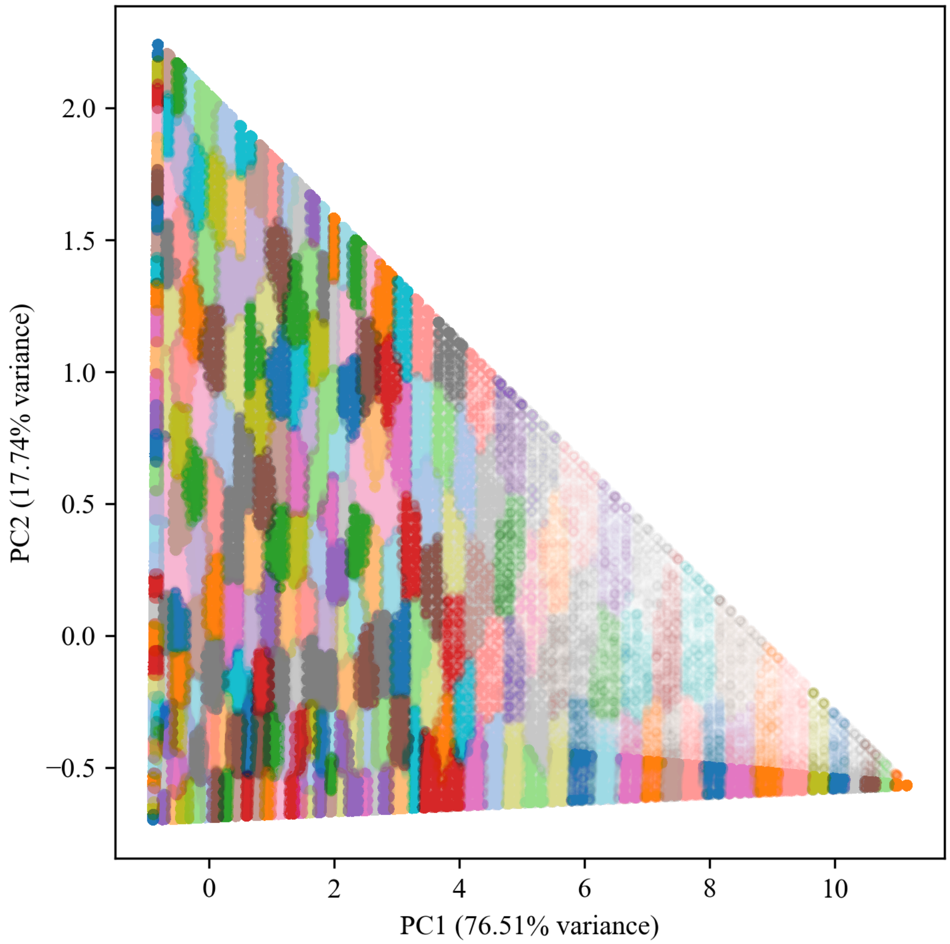

Figure 5: Clusters from which samples are draw in principal component space.

Ensuring this strategy yields high-quality training data that is representative of the study area is crucial to the success of the model. However, a rigid evaluation of the stratification approach is not possible due to the lack of ground truth data for the study area. One measure that can be taken to ensure the stratification is effective is to inspect the distribution of land cover classes across the splits. Figure **??** shows that the distribution of land cover classes is consistent across the training, validation, and testing splits, which is ideal for ensuring that the model is exposed to a representative sample of the study area's land cover diversity. One important note is that the distribution of land cover classes in the study area is highly imbalanced, with the majority of Mississippi's land cover being classified as forest, and a low density of urban and developed areas. To tackle this issue, other strategies often employ synthetic oversampling of minority classes to force the distribution of classes into a more balanced state [**?**, **?**]. Although the proposed approach to stratifying the data does not explicitly enforce such policies, comparing the land cover class distribution in the sampled tiles to the distribution across the entire state (as shown in Figure **??**) demonstrates that this sampling strategy achieves a similar outcome to such strategies. Furthermore, this approach tends to avoid sampling tiles that are composed of a single land cover class. In the original NLCD data, 29.96% of the tiles are composed of a single land cover class, while only 1.90% of the sampled tiles are composed of a single land cover class. This is promising, as heterogeneous tiles with a mixture of land cover classes convey information about the spatial organization of land cover classes. Additionally, a visual inspection of the location of the sampled tiles in Figure **??** shows that the tiles are distributed across the state of Mississippi without any apparent spatial bias.

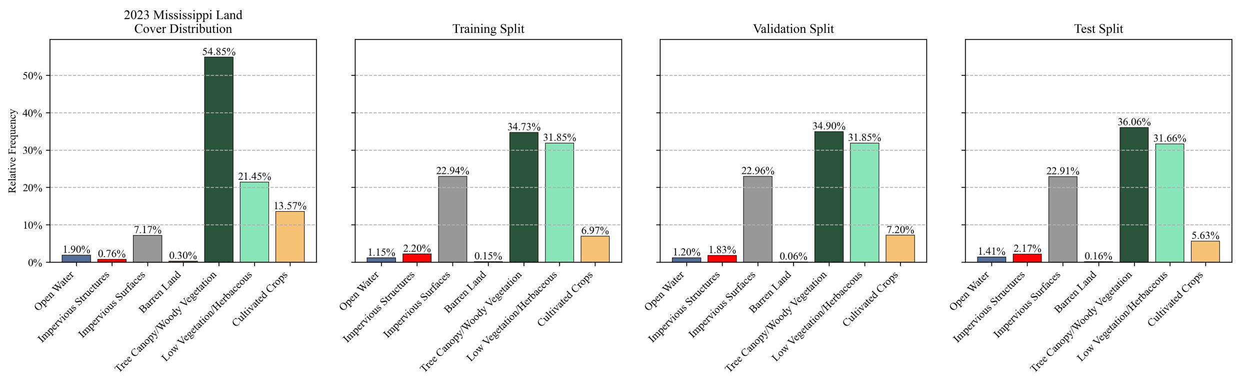

Figure 6: Distribution of 2023 NLCD land cover classes across Mississippi: statewide distribution compared with training, validation, and test splits.

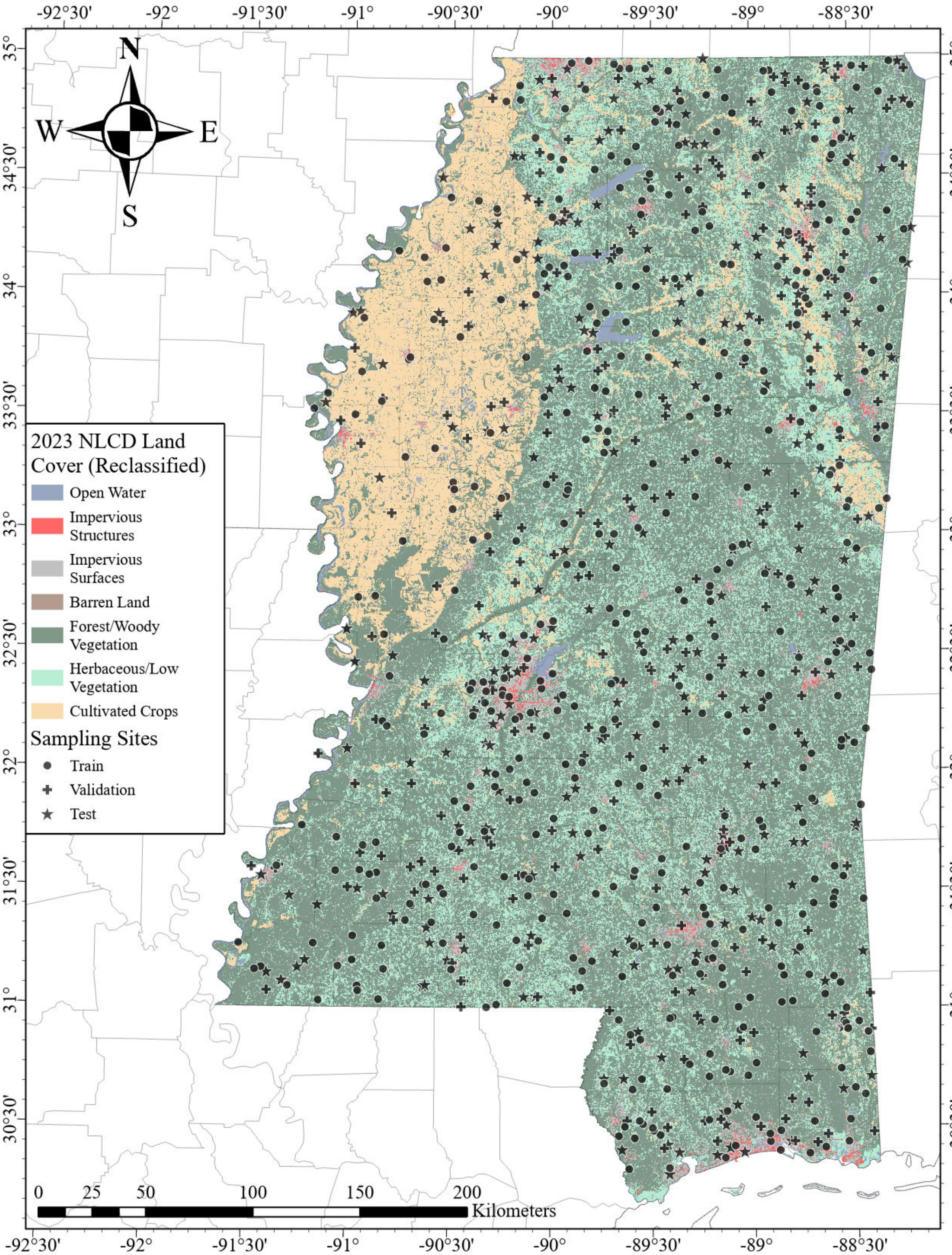

Figure 7: Sample locations for the Mississippi dataset.

3.3 Residual U-Net model configuration

The proposed model is based on the U-Net architecture, which is a popular choice for semantic segmentation tasks due to its relatively simple yet effective design and its popularity in the remote sensing community [?, ?, ?, ?]. Several modifications are made to the original U-Net architecture to improve the model's performance. First, in lieu of the original convolutional encoder network in the original implementation, a much deeper ResNet-152 network is used in its place. The reasoning behind this choice is that the increased number of layers in the ResNet-152 network facilitate more complex feature extraction, particularly during the self-supervised pre-training phase, where the large amount of unlabeled data will allow the encoder to learn to capture rich representations of spatial features in the input imagery. The number of channels accepted and produced by the decoder convolutions are subsequently adjusted to match the output of their corresponding encoder blocks. Second, the transposed convolutional layers in the original U-Net decoder are replaced with bilinear upsampling to reduce the presence of checkerboard artifacts in the upsampled feature maps, which can inhibit the quality of the output segmentation maps [?, ?]. Finally, in each decoder block, residual connections are used to facilitate the flow of information through the network and improve training stability [?].

3.4 Multi-stage pre-training approach

The proposed pre-training approach consists of two phases: a self-supervised pre-training phase and a fully supervised pre-training phase. As shown in Figure **??**, the model is first trained using the unannotated NAIP imagery over the state of Mississippi, then additional pre-training is performed using NAIP imagery over Fairfax County, Virginia using ground truth land cover data from the Chesapeake Bay Land Cover Dataset, before being fine-tuned on our hand-labeled training data from Mississippi. The purpose of such a pre-training regimen is to leverage the large volume of imagery available to train the encoder network to extract features that are useful for image reconstruction as well as relevant for land cover classification [?]. The second phase of pre-training encourages the model to refine these features in a supervised setting, while also initializing the decoder network using a land cover classification task that is nearly aligned with the downstream Mississippi land cover classification task [?, ?]. The goal of such an approach is to reduce the amount of labeled data required to achieve optimal performance on the final land cover classification task [?, ?, ?, ?].

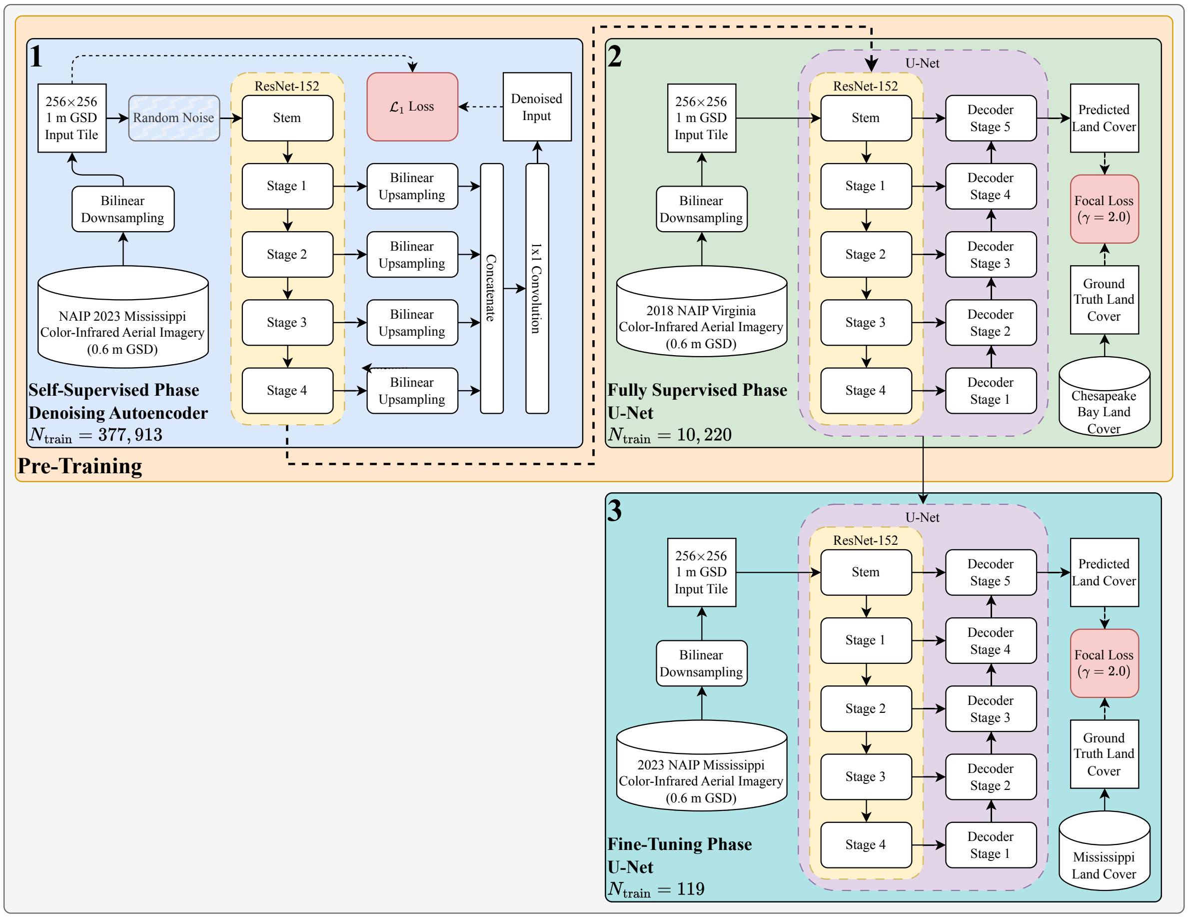

Figure 8: Flowchart of the model training workflow.

Self-supervised pre-training In the first stage of pre-training, a denoising autoencoder is trained using the unannotated NAIP imagery over the state of Mississippi. This denoising autoencoder consists of a ResNet-152 encoder network and a simple decoder network that receives feature maps from each stage of the encoder network, resamples them to the original input resolution using bilinear interpolation, concatenates them, and passes them through a single 1×1 convolutional layer to produce an output with the same dimensions as the input. The ResNet-152 encoder network is initialized with ImageNet-1k pre-trained weights, while the decoder network is randomly initialized. A full decoder network (as is common in traditional autoencoders) is not used in this phase, as the primary goal of this stage of pre-training is to force the encoder network to learn robust features present in the imagery that are necessary for reconstruction, rather than to necessarily learn to denoise the input images with a high degree of accuracy. The single-layer decoder network is designed to ensure that backpropagation primarily influences the parameters of the encoder network, while also reducing the computational cost of pre-training the model.

Two types of noise are added to the image: Gaussian noise and Poisson noise. The intensity of the Gaussian noise is controlled by the standard deviation parameter σ, which is randomly sampled from a uniform distribution $U(0, 1)$. Similarly, the intensity of the Poisson noise is controlled by the rate parameter λ, which is also randomly sampled from a uniform distribution $U(0, 1)$. The model inputs are standardized to have zero mean and unit variance on a per-band basis using the mean and standard deviation of the pre-training data prior to adding noise. The model is trained using Layer-wise Adaptive Rate Scaling (LARS) optimizer in order to stabilize training with a large batch size of 1024 [?]. After each epoch of pre-training, the model is evaluated using the pre-training validation set to ensure that the model is fitting the data properly and to adjust the learning rate if necessary. If the validation loss does not decrease after 5 epochs, the learning rate is reduced by a factor of 0.1. If the validation loss does not decrease after 15 epochs, the training is terminated early.

Fully supervised pre-training Once the self-supervised pre-training phase is complete, the encoder network is used as the encoder in the U-Net architecture, and the decoder network is initialized with random weights. The model is trained to minimize the Focal loss, defined as:

$$\mathcal{L}_{\text{Focal}}\left(p_c\right) = \begin{cases} -\alpha_c\left(1-p_c\right)^{\gamma}\log\left(p_c\right) & \text{if } y = c \\ -\left(1-\alpha_c\right)p_c^{\gamma}\log\left(1-p_c\right) & \text{otherwise} \end{cases} \tag{1}$$

where p_c is the predicted probability of class c, y is the true class label, α_c is the class weight for class c, and γ is a hyperparameter that controls the degree of class weighting. In order to address the class imbalance present in the data, class weights are assigned according to the following formula:

$$\alpha_c = \frac{C\left(1-\rho_c\right)^{\beta}}{\sum_{c=1}^{C}\left(1-\rho_c\right)^{\beta}} \tag{2}$$

where ρ_c is the proportion of class c in the training set, β is a hyperparameter that controls the degree of class weighting, and C is the total number of classes. γ is set to 2.0 in accordance with the authors' findings in the original Focal loss implementation [?], and $\beta = 2.0$ such that minority classes are emphasized without majority classes being underrepresented in the loss function. The model's parameters are optimized using the AdamW optimizer with a batch size of 64 and an initial learning rate of 1e-4. The model is evaluated after every epoch using the validation set to adjust the learning rate and terminate the training early if necessary: if no improvement is observed in the validation loss for 3 consecutive epochs, the learning rate is reduced by a factor of 0.1, and if no improvement is observed for 15 consecutive epochs, the training is terminated early. During this phase, the model is trained using two different configurations: one where the entire model is trained end-to-end, and one where only the decoder network is trained while the encoder network is frozen. Three different sets of encoder parameters are also experimented with: ImageNet-1k pre-trained weights, randomly initialized weights, and weights pre-trained using the self-supervised pre-training phase. After each model has been trained, they are evaluated using the withheld testing set to examine the performance of each training configuration on the Fairfax County land cover classification task. Evaluating the model on this task, while not directly related to the downstream Mississippi land cover classification task, gives an indication of the model's ability to learn relevant features from the data and to generalize to a new, but similar, land cover classification task.

Fine-tuning During the final fine-tuning phase, the model is trained using the labeled data from the state of Mississippi. The same loss function, class weighting strategy, optimizer, and hyperparameters as the fully supervised pre-training phase are employed. Several sets of initial weights are tested, including random initialization, ImageNet-1k pre-trained weights in the encoder network, and weights pre-trained using the self-supervised and/or fully supervised pre-training phases. Further, for each model configuration, three different scenarios are tested: one where the entire model is trained end-to-end, one where the encoder parameters are frozen and only the decoder is trained, and one where the entire model is frozen and only the final classification layer is trained. After training, the model is evaluated using the holdout testing set to examine the model's efficacy on the Mississippi land cover classification task.

3.5 Wide area inference for large-scale land cover mapping

Once the best performing model has been selected based on the testing set performance (specifically the macro F1 score), the model is integrated into an inference pipeline to generate land cover classification maps for the entire state of Mississippi. As the NAIP imagery is split into county-level subsets (see Section **??**), each county is processed individually, then the resulting land cover classification maps are stitched together to create a single, seamless land cover classification map for the entire state. Prior to performing inference on a county, the county's boundary is obtained from the US Census Bureau's TIGER/Line Shapefiles, buffered by 1.5 times the model input size to ensure sufficient context for accurate predictions near the county boundary, and the NAIP imagery is cropped to the buffered boundary. A "sliding window" approach is applied to inference, where the model processes overlapping patches of the cropped NAIP imagery, and the resulting probabilities are post-processed and averaged to create a single land cover classification map for the county. For the inference pipeline, a model input size of 256×256 pixels and a stride of 128 pixels are selected, which allows for a 50% overlap between adjacent patches. To reduce the impact of edge effects on the stitched land cover classification map, the probabilities of each pixel are weighted by their distance from the center of the tile by multiplying the probabilities by a 2D Gaussian kernel with the same dimensions as the model input size with $\sigma = 128$. This ensures that the probabilities of pixels near the center of the tile have a greater impact on the final land cover classification map than those near the edges of the tile. A weighted average of the probabilities of each pixel with the same coordinates in each tile is performed to create the final land cover classification map for the county.

3.6 Model evaluation and accuracy assessment

To evaluate the performance of the proposed model, several metrics will be utilized: overall accuracy, producer's accuracy, user's accuracy, F1 score, and Cohen's Kappa. Due to the severe class imbalance present in the dataset, the model performance will be measured on both a per-class basis (using macro averaging) and a per-pixel basis. The model exhibiting the highest macro-averaged F1 score on the testing set will be selected to generate the land cover classification map for the entire state of Mississippi. A thematic accuracy assessment of the resulting land cover classification map will be conducted using a random sampling approach, selecting 10,000 points across Mississippi. Regions previously used as part of the pre-training, fine-tuning, validation, or testing sets will be excluded from this accuracy assessment to ensure evaluation on unseen data. Additionally, to reduce the impact of spatial autocorrelation on the accuracy assessment, a minimum distance of 256 m will be enforced between sampled points.

4 Preliminary Results and Discussion

To collect preliminary results, 250 tiles from the full Mississippi land cover dataset were used for fine-tuning training, validation, and testing in the last stage of the multi-stage pre-training approach. In total, 119 tiles were used for training, 61 tiles were used for validation, and 70 tiles were used for testing. These early results give insight into the efficacy of this approach: Table **??** shows that introducing a denoising autoencoder pre-training stage prior to supervised pre-training has a substantial impact on a model's performance. This is evidence that the core idea of the proposed approach has merit, in that the denoising autoencoder pre-training stage enables the model to learn spatial features that are useful for land cover classification, even when the model undergoes subsequent supervised pre-training on a different region. Additionally, the choice of which layers to freeze during the fully supervised pre-training stage has a substantial impact on the model's performance. As shown in Table **??**, the most effective strategy for freezing parameters is to train the full network during the fully-supervised pre-training stage, but freeze the encoder network during the final fine-tuning stage. Examining the results of the Chesapeake Bay Land Cover classification task (i.e., the second stage of the multi-stage pre-training approach) in Table **??** shows more meager results compared to the Mississippi Land Cover classification task, with the denoising autoencoder pre-training stage providing only a marginal improvement over ImageNet pre-training. This suggests that the denoising autoencoder pre-training stage is more effective when the model undergoes further pre-training in a supervised manner. Further, this opens the door to future work on using a stronger pre-training strategy for the first stage of the multi-stage pre-training approach, such as increasing the noise level in the denoising autoencoder or using a more complex pre-text task to further improve the model's performance.

Table 4: Performance comparison of land cover classification models on Mississippi test data showing how different pre-training strategies (random initialization, ImageNet weights, and ImageNet weights followed by denoising autoencoder pre-training) affect accuracy metrics when all models undergo the same two-stage fine-tuning process: full encoder training on Chesapeake Bay data followed by decoder-only fine-tuning on Mississippi data with frozen encoders.

Stage 1 Pre-Training Strategy	Overall Accuracy (%)	Overall F1 Score (%)	Macro F1 Score (%)	Loss
Random Initialization	28.96	30.47	10.34	36044
ImageNet	77.54	77.58	50.25	19638
ImageNet + Denoising Autoencoder	**82.11**	**82.62**	**70.45**	**11364**

Table 5: Effects of training on different portions of the network during the fully supervised training stages.

Weights Trained					
Fully Supervised Pre-training		Fine-tuning			Loss
Encoder	Decoder	Encoder	Decoder	Classifier Head	
	✓			✓	38126
	✓		✓	✓	19725
	✓	✓	✓	✓	14511
✓	✓			✓	38949
✓	✓		✓	✓	**11364**
✓	✓	✓	✓	✓	11725

Table 6: Performance metrics for models trained on the Chesapeake Bay Land Cover dataset with different pre-training strategies.

Pre-training strategy	Overall Accuracy (%)	Macro F1 Score (%)	Loss
None	77.51	52.59	11056
ImageNet	82.50	67.16	7393
ImageNet + Denoising Autoencoder	**82.63**	**67.43**	**7439**

The best model from the multi-stage pre-training approach was used to classify land cover in the state of Mississippi at 1 m resolution in order to understand the strengths and limitations of the proposed approach. A visual assessment of the land cover classification results for the Mississippi dataset is shown in Figure **??**. The confusion matrix for the land cover classification results is shown in Table **??**, and the classification report is shown in Table **??**. These results show that, overall, the model is fairly accurate, even with the limited labeled data available for training. However, more data is needed to improve the model's performance on certain classes, such as cultivated crops, barren land, and open water. This suggests that, despite the promising results of the multi-stage pre-training approach, more labeled data is needed to fully leverage the feature learning capabilities of the model and improve its performance on classes that are underrepresented in the training data.

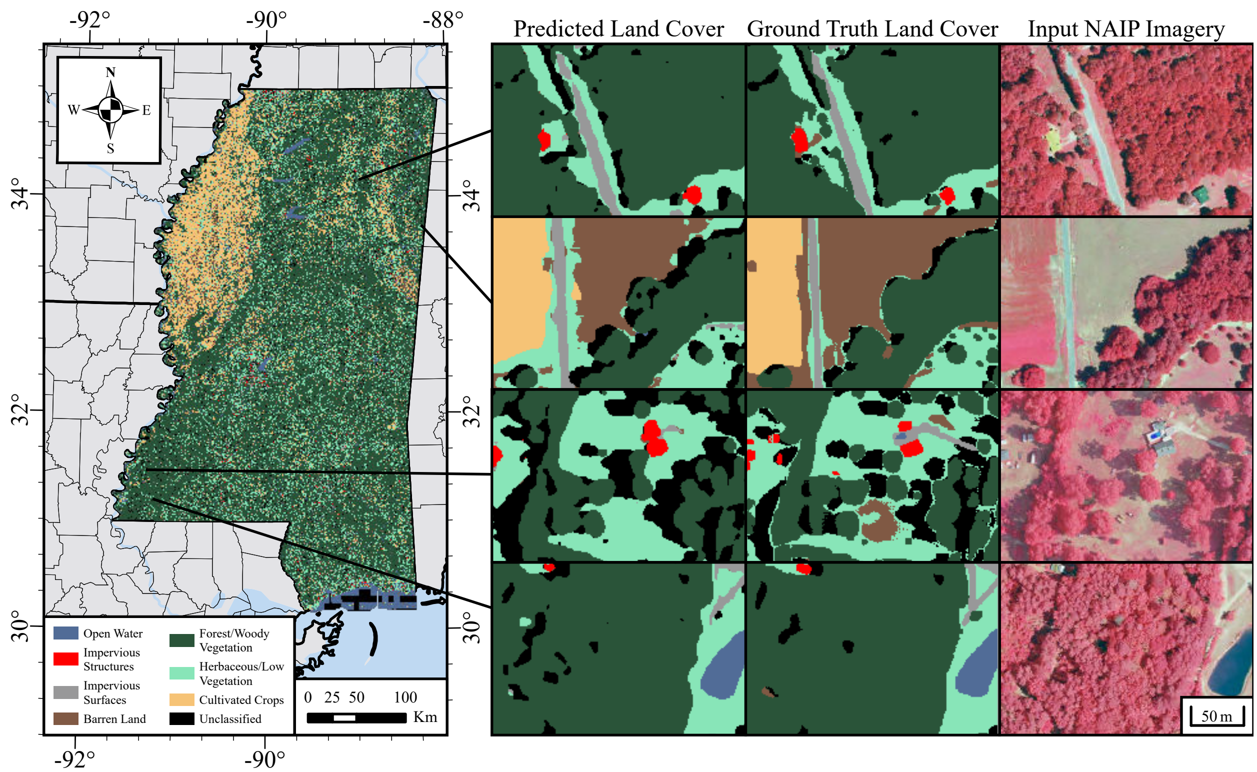

Figure 9: High-resolution land cover classification results for the Mississippi dataset (left) alongside the ground truth labels (middle) and the false-color NAIP imagery used as input (right).

Table 7: Confusion matrix for the Mississippi land cover classification results.

Predicted Label / True Label	Open Water	Impervious Structures	Impervious Surfaces	Barren Land	Tree Canopy/ Woody Vegetation	Low Vegetation/ Herbaceous	Cultivated Crops	Unclassified	Producer's Accuracy
Open Water	**7154**	344	41	45	16	141	0	1066	81.23%
Impervious Structures	13	**20776**	2041	1335	115	4891	0	1708	67.28%
Impervious Surfaces	19	2590	**124159**	9058	1722	14407	13	3225	80.00%
Barren Land	298	1964	21482	**130287**	2798	100670	24151	4008	45.61%
Tree Canopy/ Woody Vegetation	924	244	1403	1064	**2336744**	94750	26166	101005	91.20%
Low Vegetation/ Herbaceous	469	2596	24339	57498	58379	**851272**	136447	29850	73.33%
Cultivated Crops	0	0	171	3079	6685	48908	**148924**	603	54.73%
Unclassified	208	940	3546	384	14128	8236	106	**147915**	62.64%
User's Accuracy	78.75%	70.54%	70.07%	64.26%	96.53%	75.78%	44.34%	51.11%	Overall Accuracy 82.11%

Table 8: Per-class F1 scores and metrics for Mississippi land cover classification results.

Class	**F1 Score (%)**	**Support**
Open Water	79.97	8,807
Impervious (raised structures)	68.87	30,879
Impervious (paved surfaces)	74.71	155,193
Barren Land	53.35	285,658
Tree Canopy/Woody Vegetation	93.79	2,562,300
Low Vegetation/Herbaceous	74.54	1,160,850
Cultivated Crops	54.73	208,370
Unclassified	63.64	175,463
Total Support		4,587,520
Macro Metrics		
User's Accuracy (%)		68.93
Producer's Accuracy (%)		74.30
F1 Score (%)		70.45
Overall Metrics		
Accuracy (%)		82.12
User's Accuracy (%)		84.06
Producer's Accuracy (%)		82.12
F1 Score (%)		82.53
Cohen's κ		0.711

In conclusion, this research demonstrates that a multi-stage pre-training approach significantly enhances land cover classification performance under label-scarce conditions. Early results show that the proposed approach is an effective strategy for producing high spatial resolution land cover maps with limited labeled data. Beyond the immediate applications of the high spatial resolution land cover map for the state of Mississippi, this research has broader implications for land cover classification due to the unique approach to pre-training that opens the door for the development of more label efficient deep learning models. The proposed approach has the potential to enable high spatial resolution land cover classification at scale while minimizing the need for human annotation to produce ground truth data, paving the way for high resolution analyses quickly and efficiently. This is particularly important for situations where high spatial resolution land cover data for a region is necessary but existing datasets are scarce, outdated, or non-existent.

References

[1] Kumar Ayush, Burak Uzkent, Chenlin Meng, Kumar Tanmay, Marshall Burke, David Lobell, and Stefano Ermon. Geography-Aware Self-Supervised Learning. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10161–10170, Montreal, QC, Canada, October 2021. IEEE.

[2] Yifang Ban, Peng Gong, and Chandra Giri. Global land cover mapping using Earth observation satellite data: Recent progresses and challenges. *ISPRS Journal of Photogrammetry and Remote Sensing*, 103:1–6, May 2015.

[3] Thomas Blaschke, Geoffrey J. Hay, Maggi Kelly, Stefan Lang, Peter Hofmann, Elisabeth Addink, Raul Queiroz Feitosa, Freek Van Der Meer, Harald Van Der Werff, Frieke Van Coillie, and Dirk Tiede. Geographic Object-Based Image Analysis – Towards a new paradigm. *ISPRS Journal of Photogrammetry and Remote Sensing*, 87:180–191, January 2014.

[4] Thomas Blaschke, Stefan Lang, Eric Lorup, Josef Strobl, and Peter Zeil. Object-Oriented Image Processing in an Integrated GIS/Remote Sensing Environment and Perspectives for Environmental Applications. In *Umweltinformation F{\”u}r Planung, Politik Und {\”O}ffentlichkeit / Environmental Information for Planning, Politics and the Public*, volume 2, pages 555–570. Metropolis, 2000.

[5] Emmanuel Asiedu Brempong, Simon Kornblith, Ting Chen, Niki Parmar, Matthias Minderer, and Mohammad Norouzi. Denoising Pretraining for Semantic Segmentation. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 4174–4185, June 2022.

[6] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A Simple Framework for Contrastive Learning of Visual Representations. In *Proceedings of the 37th International Conference on Machine Learning*, pages 1597–1607. PMLR, November 2020.

[7] Chesapeake Bay Program. Chesapeake Bay Land Use and Land Cover (LULC) Database 2022 Edition, 2023.

[8] CVAT.ai Corporation. Computer vision annotation tool (CVAT), November 2023.

[9] Jonas Dippel, Steffen Vogler, and Johannes Höhne. Towards Fine-grained Visual Representations by Combining Contrastive Learning with Image Reconstruction and Attention-weighted Pooling, February 2022.

[10] Georgios Douzas, Fernando Bacao, Joao Fonseca, and Manvel Khudinyan. Imbalanced Learning in Land Cover Classification: Improving Minority Classes' Prediction Accuracy Using the Geometric SMOTE Algorithm. *Remote Sensing*, 11(24):3040, January 2019.

[11] Earth Resources Observation and Science (EROS) Center. National Agriculture Imagery Program (NAIP), 2017.

[12] Amanda English. Land Cover Change Analysis of the Mississippi Gulf Coast from 1975 to 2005 using Landsat MSS and TM Imagery. *University of New Orleans Theses and Dissertations*, May 2011.

[13] Jonathan R. B. Fisher, Eileen A. Acosta, P. James Dennedy-Frank, Timm Kroeger, and Timothy M. Boucher. Impact of satellite imagery spatial resolution on land use classification accuracy and modeled water quality. *Remote Sensing in Ecology and Conservation*, 4(2):137–149, 2018.

[14] Joao Fonseca, Georgios Douzas, and Fernando Bacao. Improving Imbalanced Land Cover Classification with K-Means SMOTE: Detecting and Oversampling Distinctive Minority Spectral Signatures. *Information*, 12(7):266, July 2021.

[15] George Grekousis, Giorgos Mountrakis, and Marinos Kavouras. An overview of 21 global and 43 regional land-cover mapping products. *International Journal of Remote Sensing*, 36(21):5309–5335, November 2015.

[16] Matthew C. Hansen and Thomas R. Loveland. A review of large area monitoring of land cover change using Landsat data. *Remote Sensing of Environment*, 122:66–74, July 2012.

[17] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Image Recognition, December 2015.

[18] F. Hu, G.-S. Xia, J. Hu, and L. Zhang. Transferring deep convolutional neural networks for the scene classification of high-resolution remote sensing imagery. *Remote Sensing*, 7(11):14680–14707, 2015.

[19] Brian Johnson and Zhixiao Xie. Unsupervised image segmentation evaluation and refinement using a multi-scale approach. *ISPRS Journal of Photogrammetry and Remote Sensing*, 66(4):473–483, July 2011.

[20] Reza Khatami, Giorgos Mountrakis, and Stephen V. Stehman. A meta-analysis of remote sensing research on supervised pixel-based land-cover image classification processes: General guidelines for practitioners and future research. *Remote Sensing of Environment*, 177:89–100, May 2016.

[21] Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Joan Puigcerver, Jessica Yung, Sylvain Gelly, and Neil Houlsby. Big Transfer (BiT): General Visual Representation Learning, May 2020.

[22] Nataliia Kussul, Mykola Lavreniuk, Sergii Skakun, and Andrii Shelestov. Deep Learning Classification of Land Cover and Crop Types Using Remote Sensing Data. *IEEE Geoscience and Remote Sensing Letters*, 14(5):778–782, May 2017.

[23] Adrien Lagrange, Bertrand Le Saux, Anne Beaupère, Alexandre Boulch, Adrien Chan-Hon-Tong, Stéphane Herbin, Hicham Randrianarivo, and Marin Ferecatu. Benchmarking classification of earth-observation data: From learning explicit features to convolutional networks. In *2015 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 4173–4176, July 2015.

[24] R. Li, W. Liu, L. Yang, S. Sun, W. Hu, F. Zhang, and W. Li. DeepUNet: A Deep Fully Convolutional Network for Pixel-Level Sea-Land Segmentation. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 11(11):3954–3962, 2018.

[25] Xia Li, Guangzhao Chen, Yongping Zhang, Le Yu, Zhenrong Du, Guohua Hu, and Xiaojuan Liu. The impacts of spatial resolutions on global urban-related change analyses and modeling. *iScience*, 25(12):105660, November 2022.

[26] Zewen Li, Fan Liu, Wenjie Yang, Shouheng Peng, and Jun Zhou. A Survey of Convolutional Neural Networks: Analysis, Applications, and Prospects. *IEEE Transactions on Neural Networks and Learning Systems*, 33(12):6999–7019, December 2022.

[27] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollar. Focal Loss for Dense Object Detection. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 2980–2988, 2017.

[28] Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully Convolutional Networks for Semantic Segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 3431–3440, 2015.

[29] T. R. Loveland, B. C. Reed, J. F. Brown, D. O. Ohlen, Z. Zhu, L. Yang, and J. W. Merchant. Development of a global land cover characteristics database and IGBP DISCover from 1 km AVHRR data. *International Journal of Remote Sensing*, 21(6-7):1303–1330, January 2000.

[30] D. Lu and Q. Weng. A survey of image classification methods and techniques for improving classification performance. *International Journal of Remote Sensing*, 28(5):823–870, March 2007.

[31] Lei Ma, Manchun Li, Xiaoxue Ma, Liang Cheng, Peijun Du, and Yongxue Liu. A review of supervised object-based land-cover image classification. *ISPRS Journal of Photogrammetry and Remote Sensing*, 130:277–293, August 2017.

[32] Lei Ma, Yu Liu, Xueliang Zhang, Yuanxin Ye, Gaofei Yin, and Brian Alan Johnson. Deep learning in remote sensing applications: A meta-analysis and review. *ISPRS Journal of Photogrammetry and Remote Sensing*, 152:166–177, June 2019.

[33] Yuchi Ma, Shuo Chen, Stefano Ermon, and David B. Lobell. Transfer learning in environmental remote sensing. *Remote Sensing of Environment*, 301:113924, February 2024.

[34] Oscar Manas, Alexandre Lacoste, Xavier Giro-i-Nieto, David Vazquez, and Pau Rodriguez. Seasonal Contrast: Unsupervised Pre-Training from Uncurated Remote Sensing Data. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9394–9403, Montreal, QC, Canada, October 2021. IEEE.

[35] B. L. Markham and J. R. G. Townshend. *Land Cover Classification Accuracy as a Function of Sensor Spatial Resolution*. January 1981.

[36] V. S. Martins, D. P. Roy, H. Huang, L. Boschetti, H. K. Zhang, and L. Yan. Deep learning high resolution burned area mapping by transfer learning from Landsat-8 to PlanetScope. *Remote Sensing of Environment*, 280:113203, October 2022.

[37] A.E. Maxwell, T.A. Warner, and F. Fang. Implementation of machine-learning classification in remote sensing: An applied review. *International Journal of Remote Sensing*, 39(9):2784–2817, 2018.

[38] D.K. McIver and M.A. Friedl. Estimating pixel-scale land cover classification confidence using nonparametric machine learning methods. *IEEE Transactions on Geoscience and Remote Sensing*, 39(9):1959–1968, September 2001.

[39] Mississippi State University Extension. Mississippi Land Resource Areas.

[40] Misssissippi Automated Resource Information System. Mississippi Transverse Mercator (MSTM) Projection.

[41] S.W. Myint, P. Gober, A. Brazel, S. Grossman-Clarke, and Q. Weng. Per-pixel vs. object-based classification of urban land cover extraction using high spatial resolution imagery. *Remote Sensing of Environment*, 115(5):1145–1161, 2011.

[42] Augustus Odena, Vincent Dumoulin, and Chris Olah. Deconvolution and Checkerboard Artifacts. *Distill*, 1(10):e3, October 2016.

[43] Suoyan Pan, Haiyan Guan, Yating Chen, Yongtao Yu, Wesley Nunes Gonçalves, José Marcato Junior, and Jonathan Li. Land-cover classification of multispectral LiDAR data using CNN with optimized hyper-parameters. *ISPRS Journal of Photogrammetry and Remote Sensing*, 166:241–254, August 2020.

[44] Zhuokun Pan, Jiashu Xu, Yubin Guo, Yueming Hu, and Guangxing Wang. Deep Learning Segmentation and Classification for Urban Village Using a Worldview Satellite Image Based on U-Net. *Remote Sensing*, 12(10):1574, January 2020.

[45] Darius Phiri and Justin Morgenroth. Developments in Landsat Land Cover Classification Methods: A Review. *Remote Sensing*, 9(9):967, September 2017.

[46] Rafael Pires de Lima and Kurt Marfurt. Convolutional Neural Network for Remote-Sensing Scene Classification: Transfer Learning Analysis. *Remote Sensing*, 12(1):86, January 2020.

[47] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment Anything in Images and Videos, October 2024.

[48] Caleb Robinson, Le Hou, Kolya Malkin, Rachel Soobitsky, Jacob Czawlytko, Bistra Dilkina, and Nebojsa Jojic. Large scale high-resolution land cover mapping with multi-resolution data. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12718–12727, June 2019.

[49] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation, 2015.

[50] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. ImageNet Large Scale Visual Recognition Challenge. *International Journal of Computer Vision*, 115(3):211–252, December 2015.

[51] Karshiev Sanjar, Olimov Bekhzod, Jaeil Kim, Jaesoo Kim, Anand Paul, and Jeonghong Kim. Improved U-Net: Fully Convolutional Network Model for Skin-Lesion Segmentation. *Applied Sciences*, 10(10):3658, January 2020.

[52] FL Scarpace and BK Quirk. Land-cover classification using digital processing of aerial imagery. *Photogrammetric Engineering and Remote Sensing*, 46(8):1059–1065, 1980.

[53] L. Scheibenreif, J. Hanna, M. Mommert, and D. Borth. Self-supervised Vision Transformers for Land-cover Segmentation and Classification. In *IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops*, volume 2022-June, pages 1421–1430, 2022.

[54] Hunsoo Song, Yonghyun Kim, and Yongil Kim. A Patch-Based Light Convolutional Neural Network for Land-Cover Mapping Using Landsat-8 Images. *Remote Sensing*, 11(2):114, January 2019.

[55] A.H. Strahler, C.E. Woodcock, and J.A. Smith. On the nature of models in remote sensing. *Remote Sensing of Environment*, 20(2):121–139, 1986.

[56] Mahyat Shafapour Tehrany, Biswajeet Pradhan, and Mustafa Neamah Jebuv. A comparative assessment between object and pixel-based classification approaches for land use/land cover mapping using SPOT 5 imagery. *Geocarto International*, 29(4):351–369, May 2014.

[57] Yonglong Tian, Dilip Krishnan, and Phillip Isola. Contrastive Multiview Coding. In Andrea Vedaldi, Horst Bischof, Thomas Brox, and Jan-Michael Frahm, editors, *Computer Vision – ECCV 2020*, pages 776–794, Cham, 2020. Springer International Publishing.

[58] John R Townshend. Land cover. *International Journal of Remote Sensing*, 13(6-7):1319–1328, April 1992.

[59] United States Geological Survey. Annual National Land Cover Database (NLCD) Collection 1 Products.

[60] Natural Resources Conservation Service USDA, Farm Service Agency USDA, and Rural Development USDA. Geospatial Data Gateway, 2016.

[61] Ava Vali, Sara Comai, and Matteo Matteucci. Deep Learning for Land Use and Land Cover Classification Based on Hyperspectral and Multispectral Earth Observation Data: A Review. *Remote Sensing*, 12(15):2495, August 2020.

[62] Pascal Vincent, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol. Extracting and composing robust features with denoising autoencoders. In *Proceedings of the 25th International Conference on Machine Learning*, ICML '08, pages 1096–1103, New York, NY, USA, July 2008. Association for Computing Machinery.

[63] Karl Weiss, Taghi M. Khoshgoftaar, and DingDing Wang. A survey of transfer learning. *Journal of Big Data*, 3(1):9, May 2016.

[64] Curtis E. Woodcock and Alan H. Strahler. The factor of scale in remote sensing. *Remote Sensing of Environment*, 21(3):311–332, April 1987.

[65] Zhirong Wu, Yuanjun Xiong, Stella X. Yu, and Dahua Lin. Unsupervised Feature Learning via Non-parametric Instance Discrimination. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3733–3742, June 2018.

[66] Michael A. Wulder, Nicholas C. Coops, David P. Roy, Joanne C. White, and Txomin Hermosilla. Land cover 2.0. *International Journal of Remote Sensing*, 39(12):4254–4284, June 2018.

[67] Michael Wurm, Thomas Stark, Xiao Xiang Zhu, Matthias Weigand, and Hannes Taubenböck. Semantic segmentation of slums in satellite images using transfer learning on fully convolutional neural networks. *ISPRS Journal of Photogrammetry and Remote Sensing*, 150:59–69, April 2019.

[68] Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson. How transferable are features in deep neural networks?, November 2014.

[69] Yang You, Igor Gitman, and Boris Ginsburg. Large Batch Training of Convolutional Networks, September 2017.

[70] Xiaohui Yuan, Jianfang Shi, and Lichuan Gu. A review of deep learning methods for semantic segmentation of remote sensing imagery. *Expert Systems with Applications*, 169:114417, May 2021.

[71] Xiao Xiang Zhu, Devis Tuia, Lichao Mou, Gui-Song Xia, Liangpei Zhang, Feng Xu, and Friedrich Fraundorfer. Deep Learning in Remote Sensing: A Comprehensive Review and List of Resources. *IEEE Geoscience and Remote Sensing Magazine*, 5(4):8–36, December 2017.